Cross-project quality forensics — second corpus (claude_toolkit)

Field	Value
Revi-sion	2
Cre-ated	2026-07-22T15:30:00Z
Last modi-fied	2026-07-22T16:20:00Z
Status	COMPLETE
Author	(T1/main - claude4) cross-project forensic subagent, Fable, §11.4.182
Corpus	`claude_toolkit` (operator-named; a shell/JS/Go developer-tooling repo — multi-account Claude CLI orchestration + provider-alias routing. 289 commits, 2026-05-26 → 2026-07-22) — READ-ONLY analysis, git history + files only. Nothing in the corpus was edited or executed.
Role	Second corpus for BG-QUALITY-ROOTCAUSE: cross-project validation of the reopen/bluff root-cause hypotheses on a DIFFERENT stack (bash/JS/Go tooling vs an AOSP firmware tree) with a DIFFERENT governance shape (no tracker, no gates-until-recently). Siblings: first-corpus internal forensics + `EXTERNAL_RESEARCH.md`.
Anti-bluff	Every number carries its command (§11.4.6). Every reported absence carries a control needle (§11.4.201(7)(b)). `find -newermt` was not used anywhere in this analysis (known-unreliable instrument). Unanswerable questions are marked `UNKNOWN:` with what would settle them.

. Executive summary

1. **The recurrence signature exists in full force WITHOUT any tracker machinery.** 77 of 289 commits (26.6%) are fix-typed; **38 adjacent same-scope fix-after-fix pairs landed within 24 hours of each other** — half of all fixes were followed within a day by another fix to the same subsystem. At least **11 distinct defects took ≥2 committed attempts; median attempts per multi-attempt defect = 3; maximum = 7** (one proxy-compatibility defect consumed **six consecutive releases in ~3.5 hours**, the sixth titled "complete fix").
2. **The "released fix that did not hold" root-caused (§3):** the first fix (2026-07-21) repaired the *value* (a config pin) while the *writer* — a launch path that unconditionally persists an ephemeral proxy port into a durable config with no restore-on-exit — kept re-breaking it within hours. A release titled "fully working" shipped ~12 h before the same route was measured dead in the field. The category is **value-fix vs invariant-fix**: fixing the data a defective writer produced, instead of the writer, guarantees recurrence at the frequency of the writer's execution. Even the eventual invariant-fix needed a second commit (a concurrency hardening) **2 minutes after its release**.
3. **The control-case verdict (§4) is split, and both halves matter:** the failure *causes* (green-suite-broken-product, wrong-layer testing, value-fixes, instrument false-nulls, stale-artifact shadows) appear in full without tracker machinery — so those causes are **deeper than tracker hygiene** and the first corpus's tracker-centric explanation is **incomplete**. But tracker absence removes the recurrence *signal* entirely: this project has **no reopen counter at all** — its 7-attempt chain is visible only through git archaeology, and one provider was mis-pruned on artifact grounds with nothing to reopen. The first corpus's silent-counter finding (§11.4.214) is this corpus's limit case.
4. **Governance rules existed as prose, not seams — reproduced (§5):** anti-bluff doctrine is present in all four agent-governance files, yet the repo has **zero git hooks and zero CI** (measured, control-needled), and its "MANDATORY" release gate is bound only by a comment. The gate was created **only after** a release shipped with the whole test suite green while every router alias on the real host was bricked — the project's own in-source forensic note says so verbatim.

5. **The corpus independently re-invented the constitution's mechanisms after being burned (§7):** route-attribution evidence lines, fail-closed verdicts, restart receipts (sink-side proof), a live release gate, self-validating tests (break-the-fixture-and-assert-FAIL), a schema-versioned verification cache so "results from older, weaker logic are never replayed". Convergent evolution in an unrelated project is strong evidence those mechanisms are general engineering necessities, not first-corpus idiosyncrasies.
6. **The instrument-reliability class is pervasive (§6)** — six distinct instances in the corpus's own history (swallowed `ccr restart` failures serving the wrong provider; a test runner with silent false-greens; a BRE regex-dialect guard dropping a function; SIGPIPE aborting session resolution; a `$`-token eaten out of a commit subject by shell interpolation; a changelog "Verified" count off by 111) — **plus two instances produced live by this very analysis** (§6.7), demonstrating the class's base rate even under §11.4.201 discipline.

. Corpus identity + why it is a valid control case

Commands for the identity facts:

```
$ git log --format='%h|%ad|%s' --date=iso-strict | wc -l        → 289
commits
$ git log --format=... | tail -1                                  → f83ef23
2026-05-26 "Init."
$ git log -1 --format=...                                         → a8fbbee
2026-07-22 (HEAD, branch main)
$ grep -c 'Auto-commit' <log>                                     → 17
(early era has no semantic subjects)
```

Tracker/guard machinery census (the control variable):

```
$ ls  →  docs/ qa-results/ exist;  NO Issues.md, NO Fixed.md, NO workable-items DB
$ ls .git/hooks/ | grep -v '\.sample$'   → empty (rc=1);  needle: 13 entries total incl. samples
$ ls -a .github → rc=2 (absent);  ls .gitlab-ci.yml → rc=2 (absent)
```

So: **no work-item tracker, no reopen counter, no commit-seam hook, no CI.** What it DOES have: a hermetic sandbox test suite (`scripts/tests/`, 30+ test files incl. self-validating fixtures), a proof-evidence directory (`scripts/tests/proof/`), governance files (CLAUDE/AGENTS/GEMINI/QWEN.md, each carrying anti-bluff content — `grep -c` → 2 mentions per file), and — since 2026-07-22 only — a release-gate script. This is exactly the shape needed to test whether the first corpus's failure modes require tracker machinery to occur, or only to be *seen*.

2.1. The recurrence signature measured in git

. Aggregate numbers

<table>
<tr><th>Metric</th><th>Value</th><th>Command</th></tr>
<tr><td>Total commits</td><td>289</td><td><code>git log --format='%h' | wc -l</code></td></tr>
<tr><td>Fix-typed commits</td><td>77 (26.6%)</td><td><code>grep -cE '|(fix|hotfix)[(:]' <log></code></td></tr>
<tr><td>Release-marked commits</td><td>32</td><td><code>grep -icE '|(release|v[0-9]+\.[0-9]+\.[0-9]+)' <log></code></td></tr>
<tr><td>Reverts</td><td>0</td><td><code>grep -ic revert <log></code> (needle: <code>grep -c 'un-fossilise'</code> → 2 — the instrument sees the file)</td></tr>
<tr><td>Versioned releases v1.1.0 → v1.25.4</td><td>~24 in 36 days</td><td>release list, <code>docs(release)</code> / <code>release:</code> subjects</td></tr>
<tr><td>Fix commits touching the God-file <code>scripts/lib.sh</code></td><td>38 of 77 (49%)</td><td>per-fix <code>git show --name-only</code> census</td></tr>
<tr><td>Adjacent same-scope fix→fix pairs within 24 h</td><td>38</td><td>awk over <code>(epoch, scope)</code> pairs from conventional-commit scopes</td></tr>
</table>

Per-scope recurrence (scopes with ≥2 fixes; `within24h` = adjacent same-scope pairs <86400 s apart):

```
providers      fixes=29  within24h_pairs=17
session        fixes=7   within24h_pairs=3
toolkit        fixes=6   within24h_pairs=4
(no scope)     fixes=5   within24h_pairs=3
unify          fixes=4   within24h_pairs=3
proxy          fixes=4   within24h_pairs=2
tests          fixes=3   within24h_pairs=2
scripts/toon   fixes=2   within24h_pairs=1
llmsverifier   fixes=2   within24h_pairs=1
install        fixes=2   within24h_pairs=1
cma            fixes=2   within24h_pairs=1
```

Zero reverts + 77 fixes is itself a signature: failed fixes here are never rolled back, they are **fixed forward** — which is precisely why attempt-chains, not revert-chains, are the recurrence fingerprint in an untracked repo.

```
  .    Named multi-attempt defect chains (manual
clustering, evidence = commit subjects + dates)
```

#	Defect	Attempts	Window	Chain
1	Poe proxy request/response compatibility	**7**	06-21 21:49 → 06-22 10:28 (~13 h)	tool-format → auto-start → all aliases → gzip → `$ref` schemas → "**complete fix**" (v1.7.0) → port-ready check (v1.7.1). **Six releases in ~3.5 h (06-21 21:50 → 06-22 01:21) each titled a Poe-proxy fix.**
2	ccr binary identity + launch ("is the thing I'm launching the right binary, and did it launch")	**5**	07-18 → 07-22 (4 days)	launch grammar → identity guard (v1.18.1) → migration marker for identity check → stale-bundled-ccr self-heal (v1.25.1) → PATH-shadowing doppelgänger resolve-by-stable-path (v1.25.2)
3	Session CWD-hook / cross-project hijack	**5**	07-10 19:14 → 07-11 14:05 (<20 h)	v1.13.0 fix + **four same-day follow-ups** (hook in 2nd wrapper; gate when not in worktree; migration marker; migration guard)
4	helixagent endpoint / proxy-address fossil (§3)	**3 committed** (+1 in-field failure between)	07-21 14:31 → 07-22 19:52	value-fix → invariant-fix → concurrency harden **2 min post-release**
5	CLAUDE_BIN resolution / empty-command alias	3	06-29 → 07-12 (2 weeks)	resolve across npm/brew → self-heal rc lines → self-heal in both wrappers (v1.13.2)

#	Defect	Attempts	Window	Chain
6	Output-token clamp	3	07-18 14:20 → 07-19 02:34	cap (v1.16.0) → clamp ≤128000 + isolate guards (same day) → test hardening (v1.18.1)
7	Test-runner false-greens	2	07-19, 6 min apart	"unmask 5 hidden failures" → "harden runner against silent false-green"
8	Orphaned provider records	2	07-19, 45 min apart	prune → distinguish the two orphan classes
9	toon byte-parity	2	07-22	review fix → re-review fix
10	llmsverifier auth/compat	2	06-16	auth-header patch → coherе compat + fallback
11	install rc-line hygiene	2	06-29	resolve → self-heal dangling/duplicate lines

≥11 distinct defects took ≥2 committed attempts. Attempt distribution {7,5,5,3,3,3,2,2,2,2,2} → median 3. Clustering is conservative (subjects + file overlap only); the true count is a lower bound. `UNKNOWN:` recurrences inside the 17-commit "Auto-commit" era (2026-05-26 → early June) — subjects carry no information; settling it would require per-commit diff archaeology, out of proportion here.

.   Release-to-recurrence latency

Release claim	Recurrence evidence	Latency
v1.25.0 (07-22 01:55) "…HelixAgent **fully working**…"	v1.25.1 (07-22 12:42): "router aliases refused to launch"; the gate's in-source forensic note: "v1.25.1 shipped with the whole sandbox suite green while **EVERY router alias on the real host was bricked**"	**~11 h**
v1.25.4 (07-22 19:50) "un-fossilise… (kill 502 class)"	a8fbbee (07-22 19:52) hardens the fix's own marker r-m-w against lost updates	**2 min**
v1.6.4 (06-21 21:50) "Poe proxy fix"	v1.6.5 (06-21 22:43) "Poe proxy fix"	**53 min**
v1.6.9 (06-21 23:46)	v1.7.0 (06-22 01:21) "Poe proxy **complete** fix"	95 min
v1.7.0 "complete fix"	v1.7.1 (06-22 10:28) port-ready check for the same proxy	~9 h
f72a756 (07-21 14:31) helixagent pin fixed	route re-fossilised in the field by 07-21 16:36Z (first-corpus DIAGNOSIS §3.4, watchdog log)	**~7 h**

The latency distribution is **minutes-to-hours, not weeks** — meaning in this corpus recurrence was not a slow decay but an immediate consequence of fixing the wrong layer. The word "complete" appearing in the 6th attempt of a chain is the linguistic form of the same claim-vs-reality gap the first corpus measures with reopen counters.

. Case study: the released fix that did not hold (the fossil defect)

This is the single most instructive object in the corpus: a defect **fixed, released, and measured still live in production on first touch** — reproduced in a second, independent project, with the whole causal chain recoverable from git.

. Chronology (all timestamps + ; commands: `git log -1 --format='%h %ad %s' <hash>`, plus the first-corpus DIAGNOSIS measurements of - - ~ : - :)

When	Event
07-21 14:31	`f72a756` **fix #1 (value-fix):** the provider pin pointed at the gateway it was supposed to route *through* (self-defeating config). Fix: repoint the stored `base_url` at the real backing server. The commit message itself documents the pre-gate era bluff: *"Before those gates existed it was badged* `verified` *on turns actually served by whichever provider ran last."*
07-21 ~16:36Z	The launch path re-stamps the route; the durable config again names an **ephemeral proxy port** owned by an already-exited launch (first-corpus DIAGNOSIS §3.4, watchdog log census: 103 ticks on the dead route).
07-22 01:55	`5ac6a49` release v1.25.0: "…**HelixAgent fully working**…"
07-22 12:42→16:33	v1.25.1, v1.25.2, v1.25.3 — three more same-day fix-releases in the same blast radius (stale bundled binary; PATH-shadowing doppelgänger; a setting lost on generator regen).
07-22 ~16:33	**Field measurement (first corpus):** `Router.default` names `127.0.0.1:3457` — dead (curl `000`); the real backend is alive on `:18434` (curl `200`). A *second* provider (`poe`) carries the identical fossil, and its `502` had been recorded as a **provider outage** — it was pruned from a rotation list on artifact grounds.
07-22 19:08	`0382019` **fix #2 (invariant-fix):** CAS exit-repair + crash-safe stale-reap — the durable file may never outlive the ephemeral thing it names.
07-22 19:50	`18780db` release v1.25.4 "kill 502 class".
07-22 19:52	`a8fbbee` **fix #3:** hardens fix #2's own marker read-modify-write — it had a lost-update race across three un-serialised writers plus a word-split bug. Its body: the four findings came from fix #2's code review, were classified *non-blocking*, and were landed **2 minutes after the release**.

. Why the released fix did not hold — the root cause, stated generally

`f72a756` fixed the **data** (the stored endpoint value). The **writer** remained defective in four independent, compounding ways (all proven in the first-corpus DIAGNOSIS §§1–3 against `scripts/lib.sh`):

1. **Ephemeral-into-durable:** the launch path persisted a scan-until-free ephemeral proxy port into a durable config file. The persisted value was not even reproducible across launches, let alone valid after exit (§11.4.111's resolve-by-stable-identity violated at the *lifetime* dimension, not just the naming dimension).
2. **Unconditional overwrite, last-launcher-wins:** plain jq assignment, no guard, no restore-on-exit — the route outlives the launch by design.
3. **Swallowed failure in the apply step:** `"$_ccr" restart >/dev/null 2>&1 || true` (`lib.sh:1580`) — a refused restart leaves the *previous* provider serving **while the config file reads back correct** (the project's own CLAUDE.md documents this consequence verbatim).
4. **No seam checked the invariant:** nothing asserted "the endpoint the durable config names is currently listening" — so the fossil was invisible to every green run until an end-to-end field probe hit it.

General law this instantiates: *a fix that repairs the artifact a defective writer produced, without repairing the writer, has a guaranteed recurrence horizon equal to the writer's next execution.* Here the writer ran on every alias launch, so the fix held for hours. The first corpus's "fixed-but-broken-on-first-touch" items and this corpus's fossil are the same object. Corroborating same-day siblings *inside this corpus*: v1.25.3 exists because a setting introduced in v1.25.2 **did not survive the generator regen** (same class: durable intent lost to a regenerating writer), and v1.25.1 exists because a **stale built artifact** shadowed the fixed source (SOURCE ≠ ARTIFACT, §11.4.108 layer 2).

. And the invariant-fix itself needed a fix

Fix #2 introduced a marker file maintained by read-modify-write at three call sites with no mutual exclusion; review found the lost-update race pre-release; it was classified non-blocking; the release shipped; the harden landed 120 seconds later. Two lessons, both generic: (a) **a "harden X" commit immediately after "fix X" is**

a silent reopen even when no tracker exists to record it; (b) review findings on a fix's *concurrency* are not "non-blocking nits" when the fix's whole purpose is state integrity (§11.4.194's assumed-away-factor pattern, at severity one notch lower).

. Attribution note (§ . .)

The task framing "fixed, released, and still broken on first touch" compresses two fixes into one: the *committed-before-the-measurement* fix that failed to hold is `f72a756` (value-fix, 07-21); `0382019`/v1.25.4 (invariant-fix) was committed ~2.5 h *after* the 16:33 field measurement and has since been live-verified (release body: "8 valid aliases PING-OK, poe fossil-cleared to real 402"). The corrected statement is stronger, not weaker: **the value-fix failed within 7 hours; a release then claimed "fully working" on top of the still-defective writer; and the field caught it before the project did.**

. The control-case answer: do the failure modes appear WITHOUT tracker machinery?

Yes for the causes; the tracker's absence removes only the signal. Both halves are findings.

- Causes present without any tracker/guard machinery (occurrence is independent of tracker hygiene)

First-corpus failure mode	This corpus, without tracker machinery	Evidence
Green suite, broken product	"v1.25.1 shipped with the whole sandbox suite green while EVERY router alias on the real host was bricked. The sandbox proves wrapper LOGIC; it is structurally blind to real-host state."	The project's own forensic comment, `scripts/claude-release-gate.sh` header
PASS attributed to the wrong thing (evidence non-attribution)	A provider "was badged `verified` on turns actually served by whichever provider ran last"	`f72a756` commit message; CLAUDE.md route-attribution doctrine
Fixed-but-broken-on-first-touch	§3 in full	measured
SOURCE ≠ ARTIFACT shadowing	stale bundled binary (v1.25.1); emitted alias file lagging `lib.sh` (first-corpus DIAGNOSIS §8.3.5 measured the deployed artifact at 0 hits while source had 4 — control-needled)	commit subjects + DIAGNOSIS
Evidence-claims drift	`8e8fb96` "docs(changelog): correct test_providers.sh count 294 → 405 in v1.24.0 **Verified** section" — a released Verified-claim off by 111	commit subject
Guard asserting a proxy signal	the 4-min watchdog required `GET / == 200` where the healthy service answers 404 → permanent false positive masking real outages (first-corpus DIAGNOSIS §5; fixed same day with golden-good/bad/negative-control fixtures)	measured
Test instrument itself bluffing	"unmask 5 hidden failures — missing summary + SIGPIPE assertions"; "harden runner against **silent false-green**"	commit subjects, 07-19

Conclusion 1: these causes are **deeper than tracker hygiene**. A project with zero tracker debt, zero unguarded-item backlog, and zero reopen statistics still produced every one of them. Any first-corpus explanation that reduces the 95%-unguarded / 52%-reopen numbers to "the tracker discipline decayed" is therefore **incomplete**: the tracker records the disease; it does not cause or cure it. The shared upstream causes visible in both corpora are (a) **testing at the wrong layer** (hermetic/sandbox/source-grep green while the deployed, real-host, runtime layer is broken), (b) **value-fixes on writer-defects**, and (c) **unreliable instruments trusted without control needles**.

. What tracker absence does cost (the signal side)

- The 7-attempt Poe chain, the 5-attempt ccr-identity chain, and the fossil chain exist **nowhere except in git subjects**. There is no reopen counter to rank fragility, so nothing steers extra scrutiny to the empirically-most-fragile subsystems (§11.4.132(d)/§11.4.189 have no input signal here at all).
- The `poe` mis-pruning (§3.1) shows the sharper cost: a *wrong verdict about a component* was acted on (removed from a rotation) **with no item to reopen when the verdict was falsified**. In a tracked project that is a §11.4.214 link+reopen; here the correction depends entirely on someone re-deriving the whole chain — which is what the first-corpus DIAGNOSIS had to do, at full forensic cost.
- **Conclusion 2:** the first corpus's silenced-reopen-counter finding (recurrences re-entering as new ids, §11.4.214) and this corpus are two points on one axis — *degraded signal* vs *no signal*. The machinery matters for **detection, prioritisation, and correction-of-verdicts**, not for preventing the underlying defect classes. Both corpora agree on that division of labour; neither contradicts it.

. What this corpus has INSTEAD of a tracker — and its measured limit

Release cadence is the de-facto item granularity here (~24 versioned releases in 36 days), and the changelog carries per-release "Verified" sections. The measured limits of that substitute: a Verified count was wrong by 111 until corrected post-hoc (`8e8fb96`), and a release titled "fully working" preceded a total field brick by ~11 h.

A changelog is a claims ledger, not an evidence ledger — it has no custody chain (§11.4.146(D3)) binding a claim to a verdict artifact, which is exactly the binding whose absence the first corpus measured as 95%-unguarded.
5

---

5 1 . Testing + governance posture: prose vs seams

. What exists

- 30+ test files under `scripts/tests/` incl. concurrency, conformance, path-shadowing, restart-selfheal, proxy, constitution tests; a proof directory with per-leg evidence files; `run-proof.sh` (6 legs).
- Genuinely good practice present: `test_constitution.sh` **breaks its own fixture and asserts the verifier fails** (golden-bad self-validation, the §11.4.107(10) pattern, independently arrived at); the route-attribution gate **fails closed** (`# FAIL: route-unproven` when the resolved route cannot be read); the verification cache is **schema-versioned** "so results from older, weaker logic
5 2 are never replayed" (the stale-oracle class, pre-empted); billing errors are classified `SKIP-QUOTA`, not FAIL (§11.4.3/§11.4.201(1) analogue).

. What binds — measured

```
$ ls .git/hooks/ | grep -v '\.sample$'      → empty (rc=1; needle: 13
total entries)
$ ls -a .github / .gitlab-ci.yml            → absent (rc=2 both)
$ grep -rln 'release-gate' scripts/ *.md    → the gate's own file,
README.md, CHANGELOG.md — no calling seam
```

Nothing mechanically binds any of it. The suite runs when a human runs it. The release gate's "MANDATORY" is a comment: *"A release commit must not be made unless this gate exits 0"* — no hook, no CI, no wrapper refuses a release without a gate receipt. `UNKNOWN:` whether v1.25.4 actually ran the gate script — its release body cites live verification evidence ("8 valid aliases PING-OK"), which proves a live smoke of *some* form ran, but no gate receipt is committed; settling it would need the operator's shell history or a receipt convention that does not exist yet.

. The hypothesis test

The first corpus's central governance hypothesis — **"the rules existed as prose consulted by agents, not conditions checked by seams, and therefore did not bind"** (§11.4.205's forensic shape) — is **reproduced exactly** here:

- Anti-bluff doctrine is present in all four governance files (measured: 2 mentions each; CLAUDE.md carries a full page of route-attribution doctrine) — and v1.25.1 still shipped bricked with everything green.
- The seam (release gate) was created **only after** the incident, on 2026-07-22 (v1.25.2), with the incident written into its header as justification — the same rule-follows-forensics pattern by which the first corpus's constitution grew every one of its anchors.
- Critical nuance the corpus adds: **test quantity and even test quality were not the gap** — the suite was extensive and partly self-validating. The gap was *layer* (hermetic sandbox vs real host) and *binding* (nothing refused the release). This sharpens the universal claim: prose → seam conversion and layer-completeness (§11.4.108's four layers) are the load-bearing variables; adding more same-layer tests is not.

. Instrument-reliability class instances

The task's standing observation — most false findings have the shape *"a token that mentions X matched as X, or an instrument's silence was read as data"* (§11.4.201(6)-(8)) — is confirmed at high base rate in this corpus:

1. **Swallowed exit status with a serving-traffic consequence:** `scripts/lib.sh:1580` — `"$_ccr" restart >/dev/null 2>&1 || true`. The project's own CLAUDE.md: a genuinely refused restart "leaves the previous

provider serving **while the file reads back correct**". `|| true` census: **77 in `lib.sh` alone** (`grep -c '|| true' scripts/lib.sh` ; 16 more in `claude-providers.sh`).

2. **The exporter adapter soft-fail sweep:** `scripts/install.sh:176` — `"$LIB_DIR/claude-export-docs.sh" || cma_warn "doc export failed (continuing)"` — the exporter is run with **no arguments** (its whole default corpus) and any failure is downgraded to a warning; a broken export is indistinguishable from a green install.
3. **The test runner itself produced false-greens:** two commits on 07-19 ("unmask 5 hidden failures — missing summary + SIGPIPE assertions", "harden runner against silent false-green") — i.e. for some prior window, an unknown number of suite runs were green-by-instrument-defect. `UNKNOWN:` how many releases that window covers; settling it needs the runner defect's introduction commit bisected against release dates.
4. **Regex-dialect loss:** `fix(migration): cma_run dropped by BRE empty-group \(\) guard` (06-29) — a BRE/ERE dialect artifact silently dropped a function during migration (the §11.4.201(7)(c) "the path is part of the instrument" class, verbatim).
5. **Pipeline-exit semantics:** `fix(session): add || true guard on head -1 pipeline to prevent pipefail SIGPIPE from aborting session resolution` (07-17) — SIGPIPE + `pipefail` aborting a healthy path; note this fix is the *correct* use of `|| true` , illustrating that the class is about **knowing which status is load-bearing**, not about banning the construct.
6. **A token eaten out of a commit subject by the shell:** `24cdb33` reads "resolve in tool schemas for Grok-4" — `cat -A` proves a double space where a `$` -prefixed JSON-schema keyword (`$ref`) was interpolated away inside double quotes at commit time. Even the repo's *history* carries interpolation loss.
7. **Two live instances produced by this analysis itself** (recorded per §11.4.6 because they demonstrate the base rate): (a) an `&&` -chained probe of `.git/hooks` returned a **false null** — the empty `grep -v sample` result short-circuited the chain and silently skipped the CI checks; re-measured with `;` -separated commands and explicit rc echoes. (b) an `awk -F'|'` field split on the git log **truncated every subject containing `||`** — the fix-list initially showed commit `1475814` as "fix(session): add " because its subject contains

`|| true`; the delimiter was a carrier. Both were caught by needling, neither reached a conclusion — but both were one unchecked step away from a wrong reported absence.

Generalisation this corpus supports: in shell-based tooling the instrument-defect base rate is high enough that *any* zero/absence result not control-needled should be treated as unmeasured (§11.4.201(7)(b) as a hard floor, not a nicety) — and delimiter/dialect/exit-status layers must be enumerated as part of the instrument, because every one of the six historical instances above returned a clean, confident, wrong answer without crashing.

7

---

7 1 . Contradictions and confirmations vs the first-corpus findings

. Confirmations (independent reproduction)

1. **Reopen-without-tracker:** recurrence chains (median 3, max 7 attempts) exist in a tracker-less project; recurrence is a property of *fix methodology* (value-fix vs invariant-fix; wrong-layer validation), not of tracker bookkeeping.
2. **Green-run-on-broken-product:** reproduced verbatim, in the project's own words, without any of the first corpus's scale factors (no 300 GB tree, no device fleet, no multi-agent parallelism) — ruling out "project scale/complexity" as the necessary cause.
3. **Prose-not-seams:** reproduced (§5.3), including the rule-follows-forensics growth pattern.
4. **Instrument unreliability as a first-class defect source:** six historical + two live instances (§6).
5. **Stale-artifact/DEPLOYED-layer gap (§11.4.108):** three independent same-day instances (stale bundled binary; regen-lost setting; emitted-alias lag).
6. **Convergent evolution of the countermeasures:** route-attribution evidence lines, restart receipts, fail-closed unknowns, live release gate, golden-bad self-validation, schema-versioned oracle cache, SKIP-vs-FAIL classification — each

independently invented here after an incident. The constitution's mechanisms
7 2 are re-derivable from first principles by an unrelated project under the same failure pressure — the strongest available evidence of their universality.

. Contradictions / corrections to the first-corpus framing (the valuable part)

1. **The tracker-centric causal story is incomplete.** The first corpus's headline numbers (95% of done-claims unguarded; 52% reopen rate) invite the reading "the tracker/guard machinery decayed, hence the bluff". This corpus falsifies the sufficiency of that reading: with **no** machinery to decay, the same causes fire at the same or higher frequency. The machinery's proven role is *signal and custody* (detection, prioritisation, verdict-correction), not prevention. Constitution consequence: anchors that only tighten tracker bookkeeping attack the symptom ledger; the prevention load rests on §11.4.108 (layer completeness), §11.4.201 (instrument integrity), §11.4.146(D3) (custody), and the writer-vs-value distinction (§7.3 below).
2. **"Fix committed" is the wrong unit of progress in both corpora, but this corpus shows it *quantitatively*:** half of all fixes were followed by another same-scope fix within 24 h. A "fix" is, empirically, a ~50%-probability hypothesis until it survives its first day in the field. First-corpus processes that mark items terminal on fix-commit + same-session green are calibrated against a coin flip.
3. **Small-project null result:** none of the first corpus's environmental suspects (host memory pressure, multi-track contention, submodule topology, agent crashes) are present here — yet the defect classes are. Environmental hardening (§12.x family) is therefore *not* the lever for this class, and forensics that stop at an environmental cause for a recurrence should be treated as
7 3 incomplete (§11.4.102 four-phase arc must reach the writer/layer/instrument question).

. Candidate universal rule distilled from this corpus (for the conductor's synthesis; NOT landed as an anchor here)

Writer-repair rule: when a defect consists of wrong **state** (config value, generated artifact, cache entry, route, marker), the fix is not the corrected state — it is the repaired **writer** plus the corrected state, plus a seam that asserts the invariant the state must satisfy ("the durable never outlives the ephemeral it names"; "the generated always carries the declared"). A state-only fix MUST be classified as a mitigation with a recurrence horizon equal to the writer's next run. Evidence: §3 (fossil, ~7 h horizon), v1.25.3 (regen loss, same-day horizon), v1.25.1 (stale artifact), plus the first corpus's fixed-but-broken batch. This composes with, but is not stated by, §11.4.108 (which spans layers of one artifact's pipeline; this spans *repetitions of the writer over time*).

---

8. Honest boundaries (§11.4.6)

1. **Clustering subjectivity:** §2.2's 11 chains come from manual clustering of commit subjects + file overlap. The mechanical adjacency metric (38 same-scope <24 h pairs) is fully reproducible; the chain count is a conservative lower bound, not an exact census.
2. **Scope-token granularity:** conventional-commit scopes are coarse (`providers` covers many features), so `within24h_pairs=17` for that scope over-counts *unrelated* adjacent fixes and under-counts cross-scope recurrences of one defect. The two metrics (mechanical + manual) bracket the truth from opposite sides.
3. **`UNKNOWN:` items** — carried inline: gate execution for v1.25.4 (§5.2); false-green runner exposure window (§6.3); Auto-commit-era recurrences (§2.2); and, inherited from the first-corpus DIAGNOSIS, the identity of the process that performed the 16:33 fossil-stamping launch.
4. **Survivorship:** git shows only defects that received fix commits. Defects observed but never fixed, or never observed, are invisible to every number here; nothing in this report estimates their volume.
5. **One-project control:** this is a single control case. It falsifies the *sufficiency* of the tracker-decay explanation; it cannot by itself establish frequency distributions across the population of projects — that is the external-literature sibling's job.
6. **Read-only compliance:** no file in the corpus was modified; no corpus script was executed; all evidence is from `git log/show/grep` , `ls` , `stat` , `cat -A` , `head` , `wc` , `awk` over read paths. The two in-flight uncommitted work

streams in the corpus working tree (`scripts/lib.sh` anti-fossil deploy state, `scripts/toon/*`) were left untouched (`git status --porcelain` observed, not altered).

7. **`find -newermt`** was not used anywhere in this analysis.